ZERO TRUST MICROSEGMENTATION FOR AI WORKLOADS

Trust nothing. Verify everything. Grant least privilege. Assume breach.

SOURCES / ACTORS

- AI Agents
- Human Users
- Applications / Services
- CI/CD Pipelines
- Monitoring & Security

AI PLATFORM (PRIVATE CLOUD / VPC)

SEGMENT: DEV / TEST (Non-Production)

- Dev Notebooks
- Test Models
- Experimentation

No direct access to production assets

SEGMENT: TRAINING PIPELINE

- Data Ingestion
- Feature Engineering
- Model Training Jobs
- Hyperparameter Tuning

Isolated from inference and production

SEGMENT: MODEL REGISTRY

- Model Artifacts
- Model Weights
- Versions & Metadata

Read access controlled by policy

SEGMENT: INFERENCE (PRODUCTION)

- Model Inference Services
- Feature Serving
- API Endpoints

Only approved services can invoke

SEGMENT: DATA STORES

- Training Data
- Feature Store
- Model Weights
- Embeddings / Vectors

Granular access to data & weights

OPERATIONS & MONITORING

- Logging
- Metrics
- Audit
- Alerting

Read-only access via policies

ZERO TRUST POLICY & ENFORCEMENT LAYER

Identity & Context Verification
Verify identity, device, and context

Least Privilege Access
Allow only what is explicitly required

Continuous Verification
Validate continuously, not just at login

Microsegmentation Firewall / Policy
Enforce allow rules between segments

Comprehensive Logging & Audit
Log all access and enforce immutability

NETWORK INFRASTRUCTURE (ENFORCED MICROSEGMENTATION)

- Microsegmentation Gateways / Firewalls
- Service Mesh (mTLS)
- Network ACLs / Security Groups
- Private DNS / Service Discovery
- IDS / IPS / Threat Detection

EXAMPLE POLICY RULES

(Least Privilege, Explicit Allow)

- Training Pipeline CAN read Training Data (Port 443, mTLS)
- Training Pipeline CAN write Model Weights (Port 443, mTLS)
- Inference (Prod) CAN read Model Weights (Port 443, mTLS)
- Dev/Test Users CAN access Dev/Test Segment (No access to Prod)
- AI Agent (Fraud Analyst) CAN query Fraud Score API only (No DB access)

✕ All other traffic is DENIED by default

KEY PRINCIPLE

An AI agent should be trusted no more — and arguably less — than a human user.
Its behavior is less predictable and its actions are harder to monitor in real time.

HOW IT WORKS

1. Verify
Every request (human, agent, or service) is authenticated and context is verified.

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2. Authorize
Policy engine determines exactly what the entity is allowed to access.

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3. Enforce
Microsegmentation controls enforce allow rules between segments.

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4. Monitor
All traffic and behavior is continuously monitored for anomalies.

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5. Audit & Improve
Immutable logs enable audit, compliance, and policy refinement.

BENEFITS

- ✓ Limits blast radius of compromise
- ✓ Prevents lateral movement
- ✓ Protects sensitive data and model IP
- ✓ Enforces least privilege for AI agents and users
- ✓ Improves security, compliance, and resilience